

# Training-time electrical elimination in physics-informed neural networks for electrothermal phase reconstruction

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## Abstract

A final electrical solve can repair terminal outputs while leaving inaccurate internal states unchanged. We examine how this constraint should participate in learning temperature and phase in a synthetic two-dimensional electrothermal device. A partially eliminated PINN solves the grounded electrical problem during training and uses the same face resistances for local Joule deposition. Soft PINN and interpolation controls receive the same final electrical solve. Four paired full-label fits on two separately reconstructed protocols retain the complete device-advantage criterion under three numerical references and two readers; current and power error reductions over the soft controls are 47.03-76.52% and 47.86-77.50% with the spatial reference and finer reader. Strong interpolation supplies a ranking counterexample, and matched controls do not establish added reconstruction value from the thermal/phase interior residual package. A complete second-cycle phase gap and subsequent bounded phase-correction tests retain this limitation. In a common-measure diagnostic, an observation-preserving neural correction lowers its scalar physical objective by 89.91% while increasing the raw phase-residual square 10.32-fold. Thus objective reduction and repaired ports cannot substitute for separately evaluated dynamic consistency. The results support a tested electrical-interface configuration and delimit its attribution; they do not establish neural-specific phase completion, strict two-cycle usability, formal out-of-distribution generalization or material validation.

Keywords: physics-informed neural network; electrothermal coupling; electrical elimination; missing phase observations; phase-field reconstruction

## 1. Problem and information conditions

Small potential errors can produce large contact-current and local-heating errors in a coupled phase-transition calculation. Electrical repair can restore terminal consistency without repairing the learned evolving state. We therefore ask two separate questions: whether training with the electrical constraint improves reconstruction after every method receives the same final solve, and whether the additional thermal/phase interior residuals contribute under complete and phase-gap observation conditions.

PINNs combine observations with explicit differential-equation residuals [1]. We study how electrical constraints participate in reconstructing transient temperature and phase from prescribed field observations. The task is an offline, known-model reconstruction benchmark: geometry, coefficients, initial and boundary conditions, and the complete drive are supplied. Each of the two protocols is fitted separately. Its original observation set contains 28,875 positive-time locations and 86,625 noiseless scalar labels for potential, temperature and phase. These locations constitute 1.8046875% of the  $3,200 \times 500$  positive-time carrier positions; at each observation time, 231 spatial locations cover 7.21875% of the carrier cells. Noiseless labels still inherit the carrier's numerical discretization error. These experiments do not establish a minimal sensor arrangement, zero-shot transfer or calibrated material reconstruction.

The contribution is a specific coupling interface and a matched test of its reconstruction benefit. Neural temperature and phase determine a grounded finite-volume electrical solution. The same face resistances determine local Joule deposition in the thermal balance. Providing the same final electrical solution to the soft and interpolation controls tests whether final electrical repair explains the measured differences. The existing experiments support this training configuration relative to the tested soft comparator. They do not establish a new implicit-differentiation principle or an independent benefit from the additional thermal/phase interior residuals.

Differentiable constrained learning already includes solver-in-the-loop correction [3], hybrid FEM-NN models [4], PDE-constrained layers [15], and modular implicit differentiation [5]. Related approaches enforce hard boundary maps or constrained optimization [16], prescribed integral targets [12], differential-algebraic constraints [17], and local discrete projections [18]. Here the transient  $T/\phi$  states remain neural, while the full grounded electrical subproblem is eliminated. This identifies a physical interface adaptation and its controlled comparison; it does not imply general superiority over all penalty or constrained-optimization methods.

Reduced and all-at-once PDE formulations provide the broader setting for eliminating a constrained variable versus jointly optimizing all fields [13]. Here only quasi-static potential is eliminated; the transient states remain neural.

The field network follows the smooth modified-MLP construction of Wang et al. [2]. The spatial cell-balance thermal residual is related to control-volume learning [6], while the nonconserved phase dynamics use an Allen-Cahn-type potential [7]. The device layout and electrothermal feedback are inspired by wall-cell phase-change-memory modeling [8]; its Ge-rich GST material formulation is distinct from the present dimensionless single-phase variable. The parameters below define the synthetic benchmark and are not an oxide calibration.

## 2. Two-dimensional physical object

### 2.1 Equations, geometry and known coefficients

The computational cell is  $\Omega = [-1, 1] \times [0, 1]$ , with coordinates  $x$  and  $z$  and time  $0 \leq t \leq 2.5$ . Potential  $V$ , reduced temperature  $T$  and phase fraction  $\phi$  obey

$$\nabla \cdot (\sigma(T, \phi) \nabla V) = 0. \quad (1)$$

$$T_t + L\phi_t = \alpha \Delta T - \gamma T + Q\sigma(T, \phi)|\nabla V|^2. \quad (2)$$

$$\phi_t = M(T)[\varepsilon^2 \Delta \phi - \partial_\phi W(\phi, T)]. \quad (3)$$

The conductivity, mobility and local phase potential are

$$\sigma = \exp[\beta T + \log(R)\phi^2(3 - 2\phi)], \quad (4)$$

$$M(T) = m_c + (m_h - m_c) \text{sigmoid}[(T - T_c)/w_M], \quad (5)$$

$$W = B\phi^2(1 - \phi)^2 + D(T_c - T)\phi^2(3 - 2\phi). \quad (6)$$

We use  $\alpha = 0.10$ ,  $\gamma = 4$ ,  $L = 0.05$ ,  $Q = 4$ ,  $\beta = 0.25$ ,  $R = 8$ ,  $\varepsilon = 0.04$ ,  $B = 1$ ,  $D = 6$ ,  $T_c = 0.45$ ,  $m_c = 0.5$ ,  $m_h = 5$  and  $w_M = 0.08$ . Mobility multiplies the entire phase bracket; it is not moved inside a divergence. Because  $W$  depends on the externally evolving temperature, the model is not assigned an autonomous monotone free-energy constraint.

The full top electrode is driven by  $U(t)$ ; a centered bottom segment  $|x| \leq 0.35$  is grounded. All remaining electrical faces are insulating. The top temperature is zero; the other thermal boundaries satisfy  $\partial_n T + 0.25T = 0$ . Phase satisfies  $\partial_n \phi = 0$  on every boundary. Initially  $T = 0$  and

$$\phi_0 = 0.02 + 0.01 \exp \left[ -\frac{1}{2}((x/0.18)^2 + ((z - 0.12)/0.10)^2) \right]. \quad (7)$$

These assumptions define a synthetic wall-cell benchmark. No dimensional material parameters, experimental memory operation, or oxide-specific constitutive identification are inferred from it.

### 2.2 Initial-boundary interpretation

For the initial phase in Eq. (7), at  $x=0$  on the bottom boundary, its outward normal derivative is  $-0.12 \exp(-0.72) = -0.0584102707151966$ . It is therefore not pointwise compatible with the phase Neumann condition at  $t=0$ . The reference applies the stated discrete no-flux operator during subsequent evolution; the neural representation retains the exact initial map and penalizes phase boundary residuals. We do not assume classical smooth compatibility on the closed space-time boundary.

Direct inspection of the twelve saved historical full-label calibration/L-BFGS/audit pools for the two protocols and two seeds finds no  $t=0$  boundary node and hence zero boundary quadrature mass there. This finite-pool fact is distinct from a probabilistic argument about continuous random draws. The existing Adam rule generates boundary times from the four physical windows without explicitly inserting  $t=0$ ; it does not mathematically exclude a zero random variate. The compatibility limitation is retained regardless. Its effect on the E/D\_E ranking is unknown; neither the initial state nor the boundary rule has been changed.

### 2.3 Pulse histories and observations

Each pulse rises linearly from zero to 0.72 over its first 0.05 time units, holds until local time 0.27, then falls to zero at 0.35. The original protocol starts two pulses at 0 and 1.25. The intervention changes only the second start to 1.01,

leaving the pulse shape, total time, coefficients and boundaries fixed. The input contains exactly two pulses; it is not periodically extended to create a third pulse. Advancing the second pulse by 0.24 preserves its phase relative to the 0.02 observation spacing.

Support trajectories use  $80 \times 40$  cells, a main time step of 0.0025 and saving every two steps. The spatial and temporal masks take indices 0,4,8,... and the last index, fixed independently of field values. The resulting  $21 \times 11 \times 126$  mask contains 231 analytic initial positions and 28,875 positive-time positions, with 86,625 scalar field labels. Training receives only these samples, coordinates and known physical laws. Dense support fields, reference fields, reference currents or heating, and sealed stress cases do not enter these training losses or within-run endpoint selection. Published reference evaluations informed subsequent research questions and interpretation; the overall development process is not claimed to be reference-blind.

The evaluation reference uses  $160 \times 80$  cells, a main time step of 0.000625 and saving every four steps, giving 1001 times. Both numerical trajectories use backward-Euler finite-volume steps with converged electrical/phase/thermal block iterations and logit Newton for phase, as specified in Supplement S1.1. Their algebraic convergence tolerances do not certify discretization accuracy. The grids serve different roles; their existence alone is not a convergence study. Each protocol has its own observations and is fitted separately. The intervention is consequently a complete-case adaptation experiment, not zero-shot transfer or a formal out-of-distribution test. The two seed values are reused across protocols for paired reporting, not counted as four independent physical cases.

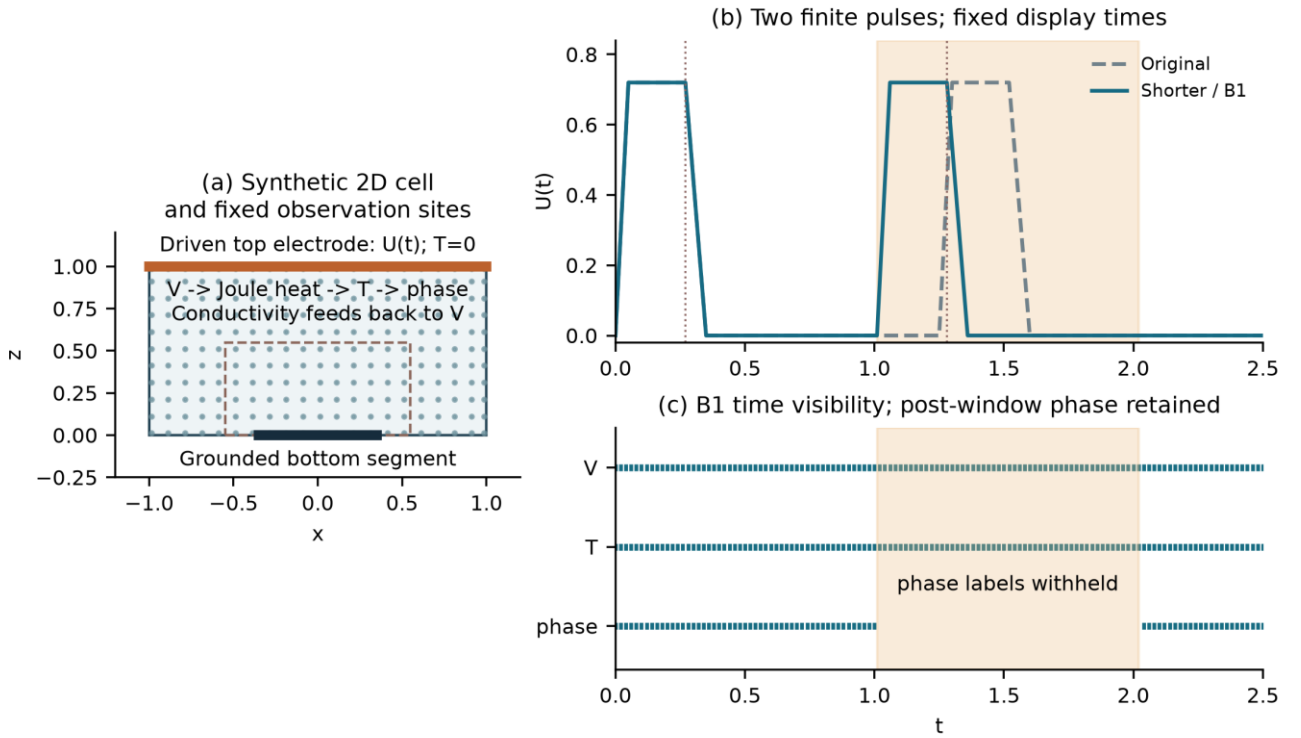


Figure 1. Actual geometry, fixed spatial sampling and finite pulse protocols. The shaded interval  $W=[1.01, 2.02]$  removes only phase labels in B1; potential and temperature remain observed. Phase observations after  $W$  remain available. Dashed vertical lines mark the predeclared display times 0.27 and 1.28. The dashed spatial box is the fixed state-error ROI.

### 3. Method and matched controls

#### 3.1 Neural states and electrical elimination

Let  $u_\theta = (T_\theta, \phi_\theta)$ . Independent smooth modified-MLP heads [2] have width 64 and four hidden layers; the shared temperature architecture also includes a small additive latent adapter described in the Supplement. FP64 arithmetic and identical output mappings are used in both methods. With  $a(t) = 1 - \exp(-t/0.35)$ ,

$$T_\theta = 2.5a(t)(1 - z)\text{sigmoid}(h_T), \quad (8)$$

$$\phi_\theta = \text{sigmoid}[\text{logit}(\phi_0) + 8a(t)h_\phi]. \quad (9)$$

The eliminated method, E, evaluates  $\sigma_\theta$  from these states and solves the grounded electrical system

$$A(\sigma_\theta)v^* = f(U, \sigma_\theta). \quad (10)$$

For an internal face  $e$  between cells  $i$  and  $j$ , the half-cell distances  $d$  and face area  $A_e$  define

$$R_{ei} = d_{ei}/(\sigma_i A_e), \quad R_{ej} = d_{ej}/(\sigma_j A_e), \quad g_e = (R_{ei} + R_{ej})^{-1}. \quad (11)$$

Electrode faces use the corresponding half-cell resistance and the actual heater overlap. Insulating faces contribute no outward conductance. Positive conductances on the connected grid, with Dirichlet electrodes fixing its gauge, make the assembled electrical matrix nonsingular. Electrical enforcement means solving this discrete system to numerical precision; it is not an exact continuum solution. The implementation uses sparse factorization, not a dense inverse. For a scalar loss, the complete first-order derivative is obtained from

$$A^T z = \partial \mathcal{L} / \partial v, \quad d\mathcal{L} = d\mathcal{L}_{\text{direct}} + z^T(df - dA v^*). \quad (12)$$

The direct term includes the explicit dependence of heating on conductivity. The right-hand side  $f$  also depends on electrode conductance, and its derivative is retained. A factorization may be reused for the adjoint of the same forward solve, but not after changing the state. This is an application of implicit differentiation [5], with no claim of a new differentiation theorem or support for arbitrary second-order derivatives.

### 3.2 Consistent local Joule deposition and remaining residuals

Given either the solved or neural potential, face current and deposition are

$$I_e = g_e(v_i - v_j), \quad P_{e \rightarrow i} = I_e^2 R_{ei}, \quad P_{e \rightarrow j} = I_e^2 R_{ej}. \quad (13)$$

Electrode power enters the neighboring boundary cell. For cell volume  $\omega_i$ ,  $q_i$  is its deposited power divided by  $\omega_i$ . This allocation is not generally an equal split between adjacent cells. It gives

$$\sum_i \omega_i q_i = P_J, \quad (14)$$

where  $P_J$  sums all internal and electrode dissipation. Equation (14) is an accounting identity for the shared face network, even when a neural potential violates electrical balance. Equality with terminal power  $UI$  additionally requires the electrical equations. Neither identity by itself establishes accuracy relative to the reference heating field.  $P_J$  denotes electrical dissipation; the reduced thermal source is  $Qq$ , so its integral is  $QP_J$ . This normalization and the temperature-dependent phase potential do not imply a general total-free-energy decay theorem.

The temperature residual uses a cell balance,

$$R_{T,i} = \partial_t(\bar{T}_i + L\bar{\phi}_i) + \gamma\bar{T}_i - \frac{\alpha}{\omega_i} \sum_f A_f (\nabla T_\theta)_f \cdot n_{if} - Qq_i. \quad (15)$$

Cell means use midpoint quadrature. Temperature gradients are differentiated at shared face nodes with opposite normals for neighboring cells, including the model's own gradients on boundary faces. The same thermal boundary residual remains in both losses. The factor  $Q$  is applied once. The phase residual is

$$R_\phi = \phi_t - M(T)[\varepsilon^2 \Delta \phi - 2B\phi(1-\phi)(1-2\phi) - 6D(T_c - T)\phi(1-\phi)]. \quad (16)$$

Thus the thermal interface is a spatial cell-balance residual and the phase interface is a pointwise AD residual. They are not claimed to equal the previous pointwise thermal formulation or the full space-time construction of [6]. At  $U = 0$ ,  $E$  uses the analytic electrical solution  $v = q = 0$  while thermal and phase evolution remain active.

### 3.3 Soft comparator and training objectives

The soft electrical PINN,  $F$ , trains all three field heads. Its potential is  $V_\theta = U(t) \exp(h_V)/[\exp(h_V)+1-z]$ , preserving the voltage range and the exact top value. It evaluates the same face operator and Joule deposition as  $E$ , but penalizes

$$r_{e,i} = (A(\sigma_\theta)V_\theta - f(U, \sigma_\theta))/\omega_i. \quad (17)$$

No additional, differently averaged electric boundary loss is added: electrical boundary conductances already enter this residual. Let  $L_{\text{obs}}$  denote the common weighted observation loss, including the full initial-logit increment for

phase, and let  $L_{BC}$  and  $L_{IC}$  be the common remaining boundary and initial losses. Their exact measures are specified in Supplement S2. With one calibration  $a, b$  at the shared parent and  $\lambda$  rising to 0.1,

$$J_{T\phi} = \frac{1}{3} \mathbb{E}_{\rho} [(R_T/4)^2 + (R_{\phi}/5)^2], \quad (18)$$

$$\mathcal{L}_E = L_{obs, E}/a + \lambda(5L_{BC} + L_{IC} + J_{T\phi, E})/b, \quad (19)$$

$$\mathcal{L}_F = L_{obs, F}/a + \lambda(5L_{BC} + L_{IC} + J_{T\phi, F})/b + \frac{\lambda\eta}{3b} \mathbb{E}_{\rho} [r_e^2]. \quad (20)$$

Here  $\rho$  is the uniform space-time target with the declared time-window quadrature. Scales  $a$  and  $b$  are shared within a pair and frozen, with a numerical floor of  $10^{-12}$ . The primary  $F$  uses  $\eta = 1$ . A development control uses a one-time gradient-balanced  $\eta$ ; a second control replaces the 128 sampled electrical cells by the volume mean over all 3200 cells at each physical time. The latter changes spatial integration, not the penalty by a factor of 25. Thermal and phase terms keep their original interfaces.

$E$  freezes the unused potential head. Its voltage observations differentiate through conductivity and the solve, whereas  $F$  voltage observations act on its potential head.  $E$  enforces its constraint at powered observation and physics times;  $F$  penalizes it at physical residual times. Initial potential, active parameters, constraint strength and gradient maps consequently differ. These are explicit parts of the method-package comparison, not hidden evidence for an isolated VJP mechanism.

### 3.4 The $D_E$ ablation and strong interpolant

$E$  learns  $T$  and  $\phi$  through the electrical solve and includes the thermal/phase interior residuals, boundary terms and initial constraints.  $F$  retains a neural potential and the frozen finite electrical penalty; its endpoint receives the identical electrical readout used by  $E$ .  $D_E$  shares  $E$ 's neural state mapping, temperature adapter, electrical solve, complete first-order conductivity feedback, voltage-observation loss, boundary/initial constraints, parent, calibration and optimization limits; only the thermal/phase interior package is removed.  $D_E$  is consequently a physics-containing ablation, not a data-only network.

$B_E$  uses time-PCHIP and spatial-linear interpolation of the supplied  $T/\phi$  observations, the original initial-logit restoration and known boundary extensions, followed by the same electrical solve. It is constructed once per protocol/information condition and does not use the  $V$  labels. The  $E/B_E$  comparison therefore measures practical competitiveness under the stated input utilization, not a strictly identical information-use intervention.

The precise ablation objective is

$$\mathcal{L}_{D_E} = L_{obs, E}/a + \lambda(5L_{BC} + L_{IC})/b. \quad (21)$$

The scale  $b$  is still calibrated from the complete parent physical package. Thus  $E$  minus  $D_E$  removes exactly the interior term rather than renormalizing the shared boundary or observation contribution. Targeted value/gradients checks and full-visible compatibility checks precede the new execution.

### 3.5 Common parents, fixed endpoints and readouts

For each protocol, seeds 29 and 43 initialize fresh networks and a zero-output adapter, without loading trained weights, optimizer history or calibration. A common observation-only parent receives 2400 Adam updates [9] and 600 full-observation L-BFGS evaluations, 200 per field head [10].  $E$  and  $F$  then start from that same fitted parent with fresh optimizers. Each branch receives 1500 Adam updates and at most 300 fixed complete-objective/gradients evaluations. Every line-search trial and repeated evaluation counts; budget termination restores the last accepted state. No reference score selects a checkpoint.

The learning rate, Adam parameters, clipping, observation stream, physical time windows and residual scales are shared. L-BFGS uses fixed data and quadrature, without resampling or gradient clipping inside its closure. Equal update and evaluation caps do not imply equal computation:  $E$  incurs sparse forward and adjoint solves, while  $F$  differentiates a potential network and its explicit residual. Actual work is retained in Supplement S6; no speedup is claimed.

The shorter-protocol  $D_E$  controls share the corresponding full-label parents.  $B1$  instead establishes two entirely new observation-only parents with only its visible phase labels, then runs  $E$ ,  $D_E$  and  $F$  in each seed without selecting a favorable checkpoint or skipping an arm based on performance. All fixed endpoints are locked before

reference scoring. The common 160 by 80 and 240 by 120 electrical readers evaluate every model at the same 1001 times. Fine-grid T/phase are direct queries of the same locked function. T/phase and event scores remain on 160 by 80; only fine potential is conservatively restricted for the inherited field guard, while native currents, power and Joule deposition stay native.

### 3.6 Distinct error and event measures

The historical primary comparison uses the same  $160 \times 80$  electrical layer at the reference times. The  $240 \times 120$  reader changes prediction-side electrical discretization while preserving the historical scoring measure. The unprojected F readout is retained separately. For a fixed F, projection changes V, q and electrical readouts while leaving T,  $\phi$  and every phase event unchanged. The error measures distinguish these effects.

S is the space-time mean symmetric difference of active sets, where active means  $\phi \geq 0.5$ ; it uses the full cell domain and global trapezoidal time weights. Raw  $E_\phi$  is phase RMS on the ROI  $|x| \leq 0.55$ ,  $0 \leq z \leq 0.55$ .  $E_T$  is ROI temperature RMS divided by 0.45, and  $E_V$  is unnormalized full-domain potential RMS. Device errors are

$$\mathcal{E}_I = \frac{\|I - I_{\text{ref}}\|_t}{\|I_{\text{ref}}\|_t}, \quad \mathcal{E}_P = \frac{\|P_J - P_{J,\text{ref}}\|_t}{\|P_{J,\text{ref}}\|_t}, \quad \mathcal{E}_W = \frac{|\int (P_J - P_{J,\text{ref}}) dt|}{|\int P_{J,\text{ref}} dt|}. \quad (22)$$

The time norm is trapezoidal RMS over the complete interval. The matched device criterion compares predicted bottom current with the reference top-current trace, retaining the historical scoring convention. The top-current noninferiority guard uses the predicted top current. For projected states these terminal traces agree to solve accuracy; the archive also reports bottom-native errors as a separate diagnostic. Unprojected soft currents can disagree substantially and must not be interchanged. Percentages below are 100 times normalized errors, not dimensional percentage errors in temperature or phase. All absolute, percentage-point and relative differences are provided in the unified tables.

For the same heating-window measure, let  $M_{\text{pred}}$ ,  $M_{\text{ref}}$  and  $M_{\text{TP}}$  be predicted, reference and overlapping active masses. When both masses exceed the numerical denominator floor, the definitions give

$$\text{recall} = \text{precision} \frac{M_{\text{pred}}}{M_{\text{ref}}} \leq \min\left(1, \frac{M_{\text{pred}}}{M_{\text{ref}}}\right). \quad (23)$$

This identity separates an insufficient predicted active mass from mismatch in space and time at a fixed mass. It is a descriptive support diagnostic, not a new loss or evidence that a particular neural mechanism caused the error. The original near-zero floors remain in the scorer. The strict gates are project-defined evaluation requirements, not experimental device-reliability standards or estimates of a success probability.

Predeclared phase-reconstruction and device-function criteria require 10% improvements in their named primary metrics with 5% noninferiority guards and fixed absolute floors. They are retained in Supplement S3, rather than used as a substitute for reporting error magnitudes. Failure to cross an advantage threshold does not demonstrate equivalence. Strict event usability is separate: both cycles must meet onset error  $\leq 0.005$ , recall  $\geq 0.9$ , precision  $\geq 0.8$ , mass ratio 0.8-1.2, and the specified peak, locality and recovery conditions. An onset is the linearly interpolated upward crossing of ROI active fraction 0.02. Support metrics integrate cell-volume overlap during each heating window.

For projected states,  $P_J = UI$  to electrical solve accuracy. Current and power errors thus differ in drive-dependent temporal weighting, but are not independent physical replications. Likewise, a small integrated energy error may reflect cancellation. We therefore retain the signed per-pulse errors, event metrics and local state errors even when global device criteria pass.

### 3.7 Frozen second-cycle phase-gap experiment

The new observation condition removes phase labels on the complete second cycle  $W=[1.01, 2.02]$  of the existing shorter protocol. V/T observations and phase outside W, including  $t > 2.02$ , remain available. The exported data contain 17,094 positive-time phase labels and omit 11,781; V and T each retain 28,875. The original observed grid contains 1.02 through 2.02 within W, with no observed node at 1.01. The last preceding and first following visible times are 1.00 and 2.04. The analytic initial condition remains known.

Two new observation-only parents, seeds 29 and 43, each initialize E,  $D_E$  and F under the frozen recipe. No complete-label trained state, calibration or interface pool is reused. Phase observation mass restricts the original coordinate quadrature before normalization. A threshold-straddling cell is admitted only when all its original

adjacent corners remain visible. The primary question is whether E adds information over D\_E in W, under the separately named A\_w criterion. Whole-history A/B, strict events, outside-window costs and E/F/B\_E comparisons remain separate. This is offline reconstruction of a developed protocol, not online forecasting or a wholly unobserved phase trajectory.

A\_w applies the original phase-advantage formula to W with its own time weights and normalizers: each of the S and phase-RMS improvements must satisfy

$$E_{\text{control}} - E_{\text{candidate}} \geq \max(0.1E_{\text{control}}, \epsilon_{\text{abs}}).$$

The absolute floor is the corresponding inherited tolerance. T, top-current and potential retain their original 5% noninferiority guards. The bottom-current convention remains separately reported. The outside measure integrates [0,1.01] and [2.02,2.5] separately, without bridging W. Its unnormalized integrals plus those of W reproduce the full-history integrals. Outside costs include all seven named state/device quantities. Original full-history A/B and strict-event rules remain unchanged. This is a complete withheld time window for phase, with V/T still observed inside it; it is not a formal out-of-distribution or forecasting test.

## 4. Reconstruction after common electrical repair

### 4.1 Four paired full-label comparisons

All four clean E/F pairs retain the complete device criterion under every existing reference and both readers. Under the spatial reference with the finer reader, current-error reductions are 47.03-76.52% (0.78-2.28 percentage points) and power-error reductions are 47.86-77.50% (0.82-2.39 points). The comparison therefore survives giving F the same final electrical solve. Four pairs on two related protocols are the comparison units; reference, reader and saved-time counts do not increase that sample size.

**Table 1. Complete full-label objects under the spatial reference and fine reader. Phase uses the original 160 by 80 ROI; native device NRMSE values are percentages. B\_E is one deterministic object per protocol. D\_E is shown alongside the learned controls rather than introduced only after the main results.**

| Protocol | Seed   | Role | Phase RMS | Current (%) | Power (%) |
|----------|--------|------|-----------|-------------|-----------|
| original | 29     | E    | 0.0210621 | 0.883264    | 0.889983  |
| original | 29     | F    | 0.0247498 | 1.66733     | 1.707     |
| original | 43     | E    | 0.0224575 | 1.59785     | 1.62156   |
| original | 43     | F    | 0.0251648 | 3.07484     | 3.18207   |
| original | shared | B_E  | 0.0233096 | 1.61121     | 1.63659   |
| shorter  | 29     | E    | 0.0199363 | 1.01558     | 1.0237    |
| shorter  | 29     | F    | 0.0244496 | 2.27681     | 2.34323   |
| shorter  | 43     | E    | 0.0197631 | 0.70046     | 0.694693  |
| shorter  | 43     | F    | 0.0240993 | 2.98378     | 3.08711   |
| shorter  | shared | B_E  | 0.0252168 | 1.9043      | 1.9475    |
| shorter  | 29     | D_E  | 0.0194683 | 0.958122    | 0.965194  |
| shorter  | 43     | D_E  | 0.0197085 | 0.642947    | 0.624503  |

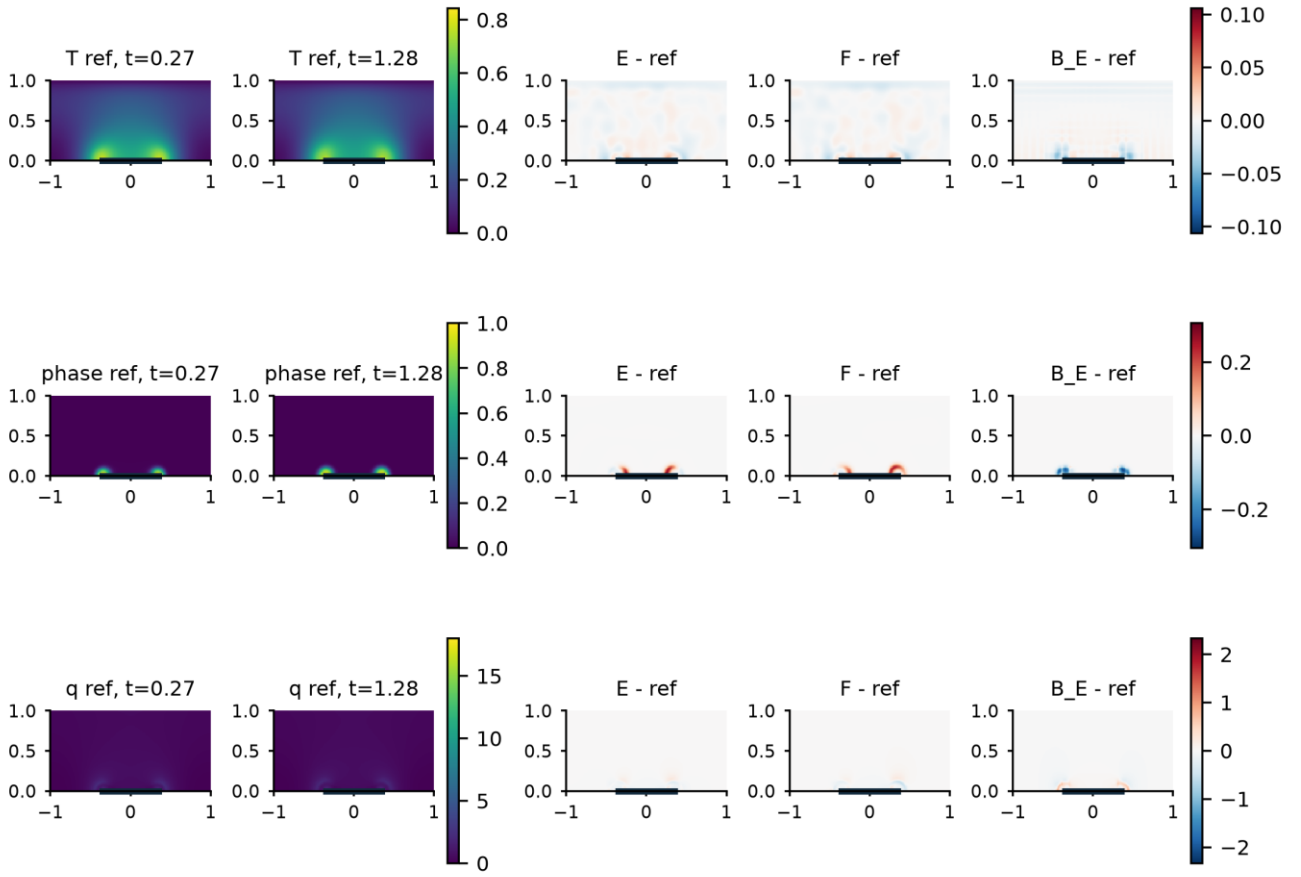
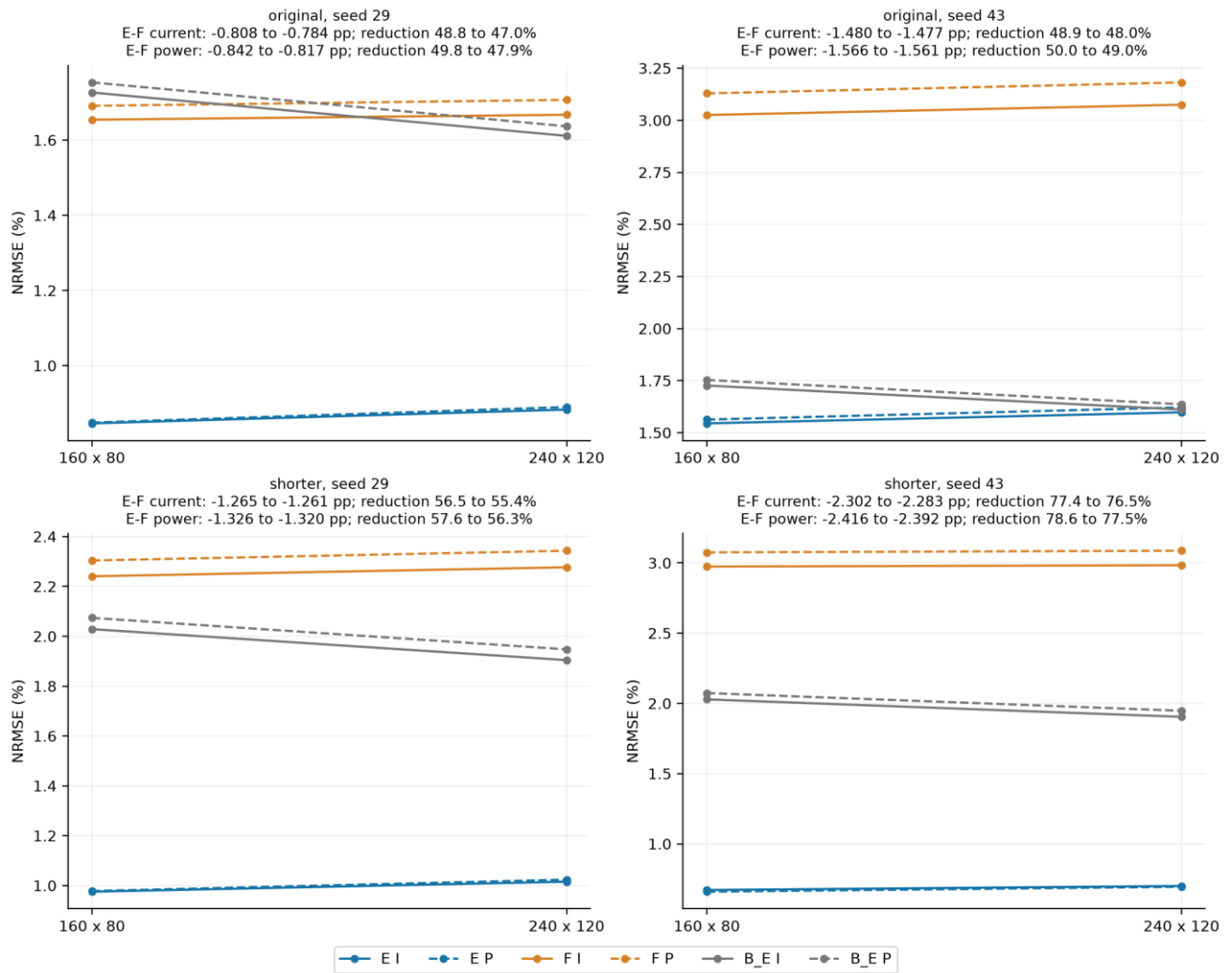


Figure 2. Native 240 by 120 physical states and signed errors for the full-label shorter protocol. The first two columns show spatial-reference T, phase and Joule density at the two predeclared plateau endpoints. The remaining columns show E, repaired F and B\_E minus reference at t=1.28 for seed 29. Error color scales are shared with the seed-43 companion in the supplement. These physical panels are descriptive native-grid views; the primary T/phase/event measure stays on 160 by 80.



**Figure 3.** All four historical E/F pairs under the same native spatial reference and the two electrical readers. Solid and dashed curves show current and power NRMSE. B\_E is repeated visually across seeds but is one baseline per protocol. Signed E-F differences use percentage points. The complete criterion also includes all field guards.

The development controls illustrate why repair is a necessary comparison: with T/phase fixed, projection reduces F<sub>raw</sub> current NRMSE from 279.30% to 1.071% and power NRMSE from 20.118% to 1.065%, while E gives 0.694% and 0.682%. A full-spatial finite-penalty control does not remove that development difference, but its integrated energy error is better. These controls delimit the tested package; they are retained in Supplement S4 and are not pooled as clean replications.

## 4.2 The strong interpolation counterexample

This robustness does not extend uniformly to B\_E. For original-protocol seed 43 and the 240×120 reader, the full-precision records give:

**Table 2.** Continuous interpolation counterexample for original-protocol seed 43 with the fine reader. Negative relative error reduction means larger E error; full-precision records determine the unchanged A/B decisions.

| Reference     | Current NRMSE: E / B E (%) | Power NRMSE: E / B E (%) | Relative current / power error reduction (%) | Original A / B |
|---------------|----------------------------|--------------------------|----------------------------------------------|----------------|
| Original      | 1.631055 / 1.620006        | 1.657409 / 1.646429      | -0.681976 / -0.666882                        | False / False  |
| Time-refined  | 1.653271 / 1.570460        | 1.681856 / 1.596088      | -5.273031 / -5.373634                        | False / False  |
| Space-refined | 1.597848 / 1.611207        | 1.621556 / 1.636592      | 0.829142 / 0.918751                          | False / False  |

Negative reduction denotes larger E error. Under the time-refined reference the differences are 0.082811 and 0.085768 percentage points, respectively. Under the spatial reference E remains slightly better, below the prescribed practical margin. The first two comparisons already failed B, so the continuous reversals introduce no additional Boolean decision changes. Full-precision source fields and line locations are preserved in [the P04 table](#); these display values do not re-adjudicate thresholds.

## 5. Dynamic residual increments and failure diagnostics

### 5.1 Full-label matched D\_E evidence

In both clean shorter-protocol pairs, D\_E has lower phase, current and power RMS errors than E under all three references and both readers. Neither pair establishes the prescribed added-residual phase or device increment. With the spatial reference and finer reader, E exceeds D\_E in phase/current/power error by 2.404%/5.997%/6.061% for seed 29 and 0.277%/8.945%/11.239% for seed 43. Other outcomes are mixed. Because the electrical solve and voltage feedback remain active in D\_E, this result concerns the additional thermal/phase interior package. It does not test physics against a data-only network. Complete independent audit components remain in Supplement S17.

### 5.2 Second-cycle phase-gap reconstruction

Neither initialization establishes the complete phase-gap criterion against D\_E under the three references and two readers. Across those conditions, relative phase-RMS reductions of E against D\_E range from -4.23% to 1.23%; negative reduction means larger E error. The prescribed phase-gap experiment does not establish the claimed additional residual increment; individual favorable errors do not replace the complete criterion.

Seed 29 has phase-RMS error reductions of 1.10% to 1.23% and active-set error reductions of 1.30% to 1.46%. Seed 43 has phase-RMS error reductions of -4.23% to -4.18% and active-set error reductions of -5.25% to -5.21%. All three window guards jointly pass in 12/12 conditions. The phase-gain requirements still determine the complete A\_w outcome; negative reductions indicate larger E errors.

**Table 3. B1 errors inside W under the spatial reference and fine reader. All seven objects are included. Full-precision values for all references/readers, including their own denominators, remain in the complete tables.**

| Seed | Role | 1000 S  | Phase RMS | T (%)   | Current (%) | Power (%) |
|------|------|---------|-----------|---------|-------------|-----------|
| 29   | E    | 8.92423 | 0.102408  | 1.14001 | 1.10488     | 1.11993   |
| 29   | D_E  | 9.042   | 0.103548  | 1.13479 | 1.15561     | 1.17082   |
| 29   | F    | 7.75023 | 0.121882  | 1.16888 | 8.96009     | 8.42207   |
| 43   | E    | 8.0082  | 0.0906255 | 1.10747 | 0.931022    | 0.95043   |
| 43   | D_E  | 7.61158 | 0.0869874 | 1.12689 | 0.996119    | 1.00989   |
| 43   | F    | 7.75023 | 0.109148  | 1.11942 | 8.90024     | 8.38301   |
| -    | B_E  | 7.75023 | 0.127808  | 1.41376 | 9.08846     | 8.55694   |

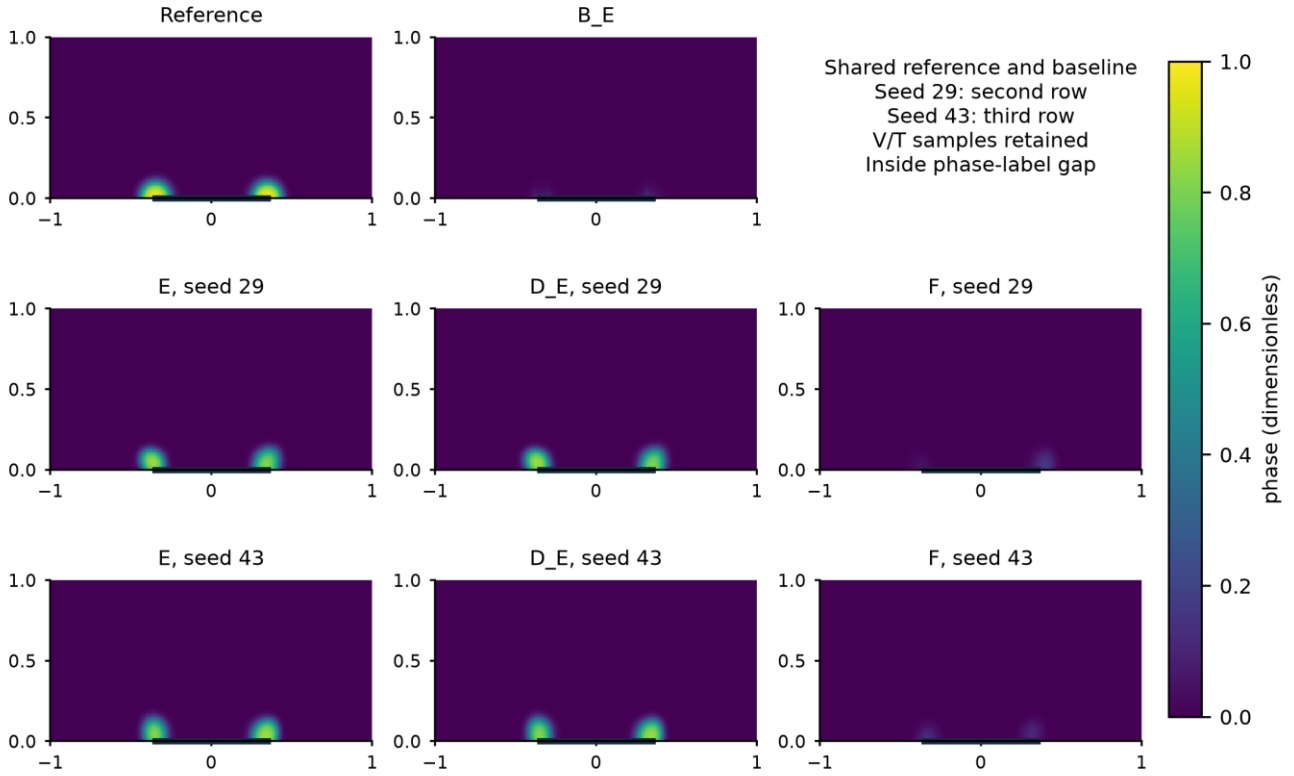
B1: phase at fixed  $t=1.28$ ; native 240 x 120 fields

Figure 4. B1 phase at  $t=1.28$ , inside the withheld second cycle. Reference and shared B\_E accompany E, D\_E and F for both seeds. All panels use the same [0,1] color range. Native fine-grid fields provide a physical view, while the primary state and event criteria use the original coarse measure. T, local q, both predeclared times and signed-error companions are retained in the supplementary physical atlas.

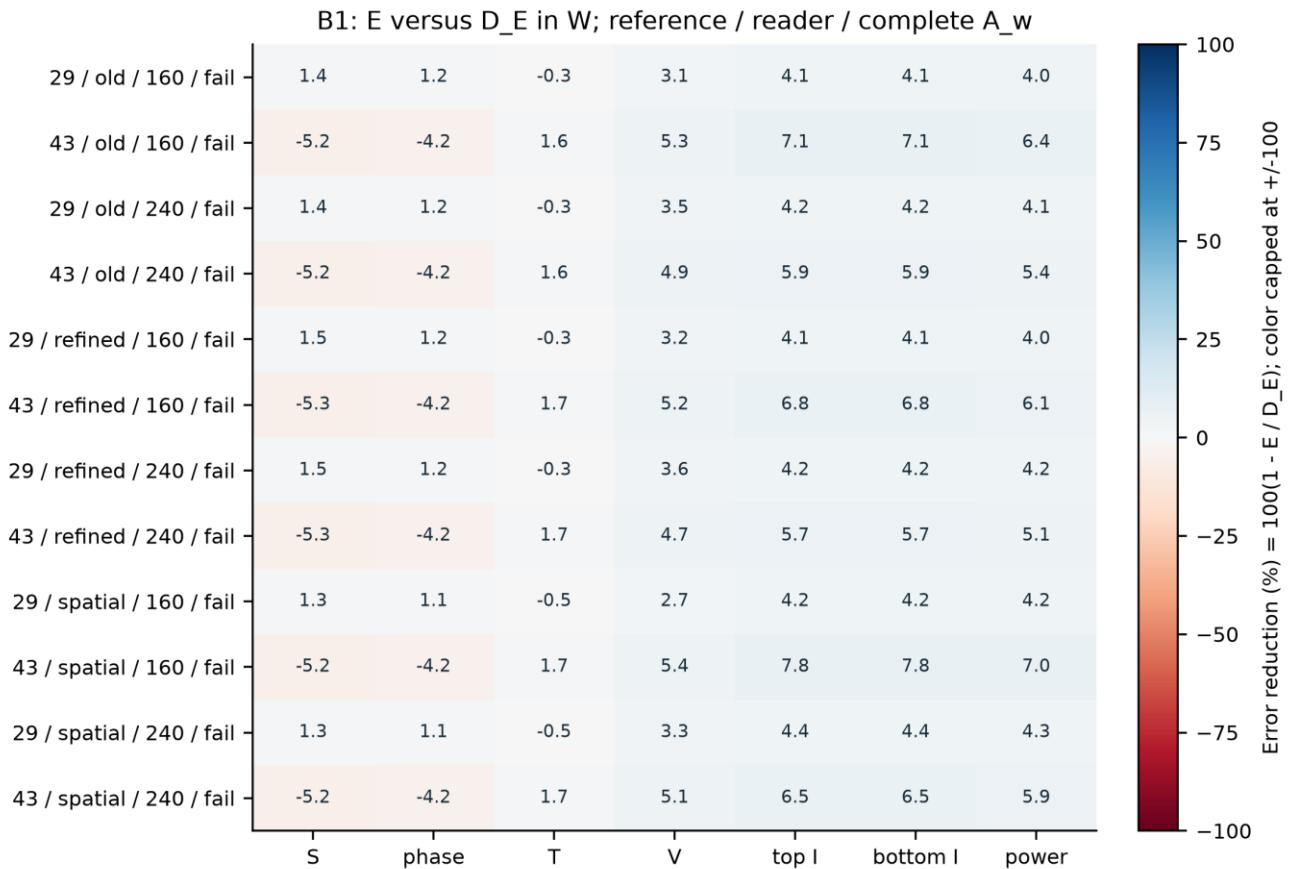


Figure 5. Window effects of E against same-parent D\_E in all twelve seed/reference/reader conditions. Positive numbers mean smaller E error. Colors are capped at plus/minus 100% for legibility; annotations retain the actual values. The row label gives the complete A\_w outcome, which also uses absolute floors and noninferiority. The rows are sensitivity checks on two independent initializations, not twelve independent experiments.

### 5.3 Whole-history costs and strict events

Outside-window noninferiority costs occur in 6 of the twelve primary comparison conditions. The affected quantities are top-current NRMSE, potential RMS, bottom-current NRMSE, power-trace NRMSE, with every condition retained in the comparison table. All seven B1 objects fail the complete strict two-cycle rule under every tested reference and reader.

**Table 4. Primary B1 decisions in every reference/reader condition. Full A/B retain their original definitions; outside cost denotes at least one separately reported noninferiority violation. A favorable window result does not erase an outside or strict-event cost.**

| Reference | Reader | Seed | A w   | Full A | Full B | Outside cost |
|-----------|--------|------|-------|--------|--------|--------------|
| old       | coarse | 29   | False | False  | False  | False        |
| old       | coarse | 43   | False | False  | False  | True         |
| old       | fine   | 29   | False | False  | False  | False        |
| old       | fine   | 43   | False | False  | False  | True         |
| refined   | coarse | 29   | False | False  | False  | False        |
| refined   | coarse | 43   | False | False  | False  | True         |
| refined   | fine   | 29   | False | False  | False  | False        |
| refined   | fine   | 43   | False | False  | False  | True         |
| spatial   | coarse | 29   | False | False  | False  | False        |
| spatial   | coarse | 43   | False | False  | False  | True         |
| spatial   | fine   | 29   | False | False  | False  | False        |
| spatial   | fine   | 43   | False | False  | False  | True         |

Against the repaired soft PINN F, B1 E satisfies the same window reconstruction formula in 0/12 conditions and the original whole-history A/B rules in 0/12 and 6/12, respectively. Against the shared interpolant B\_E, B1 E satisfies the same window reconstruction formula in 0/12 conditions and the original whole-history A/B rules in 0/12 and 10/12, respectively. These are reference/reader checks on the same two initializations. Their continuous effects and all failures remain in the complete tables; the primary attribution of the additional interior package still uses E versus D\_E. For seed 29, the full device rule passes 0/6 checks against F and 4/6 against B\_E. For seed 43, the full device rule passes 6/6 checks against F and 6/6 against B\_E.

**Table 5. Continuous B1 window effects across the three references and two readers. Each range summarizes six sensitivity conditions for one initialization; positive reduction means smaller E error. It does not establish the full window criterion.**

| Comparison | Seed | Phase reduction % | Set reduction %  | Current reduction % | Power reduction % |
|------------|------|-------------------|------------------|---------------------|-------------------|
| E / D_E    | 29   | 1.10 to 1.23      | 1.30 to 1.46     | 4.06 to 4.39        | 3.98 to 4.35      |
| E / D_E    | 43   | -4.23 to -4.18    | -5.25 to -5.21   | 5.70 to 7.79        | 5.14 to 7.04      |
| E / F      | 29   | 15.98 to 16.93    | -15.15 to -13.14 | 86.92 to 88.57      | 85.84 to 87.76    |
| E / F      | 43   | 16.97 to 18.27    | -3.33 to -1.01   | 88.69 to 90.58      | 87.70 to 89.86    |
| E / B_E    | 29   | 19.87 to 20.69    | -15.15 to -13.14 | 87.10 to 88.73      | 86.06 to 87.96    |
| E / B_E    | 43   | 29.09 to 30.23    | -3.33 to -1.01   | 88.92 to 90.78      | 87.95 to 90.07    |

The spatial-reference event records separate port and support behavior. Both E and D\_E recover a second event, whereas both repaired F states and the shared B\_E miss it. Second-cycle ROI recall is 0.7799/0.7738 for E/D\_E at seed 29 and 0.8061/0.8010 at seed 43, below the 0.9 requirement. E second-event timing errors are 0.00450 and 0.00771, respectively. Thus substantially improved window ports and recovery of an event can coexist with failure of the phase and strict two-cycle criteria. The complete first/second-event records, including support extent, precision and recovery, are retained in Supplement S20.

All E/F and E/B\_E comparisons, complete cycle metrics, signed device effects and separate full/outside normalizers are retained with the primary comparison. Phase observations after 2.02 support both the networks and the legal interpolant, so even a favorable result is offline reconstruction rather than online second-pulse prediction. The outcome has not changed the full-label D\_E finding, historical E/F decisions, seeds, thresholds or reference definitions.

## 5.4 Bounded corrections and equation-wise diagnostics

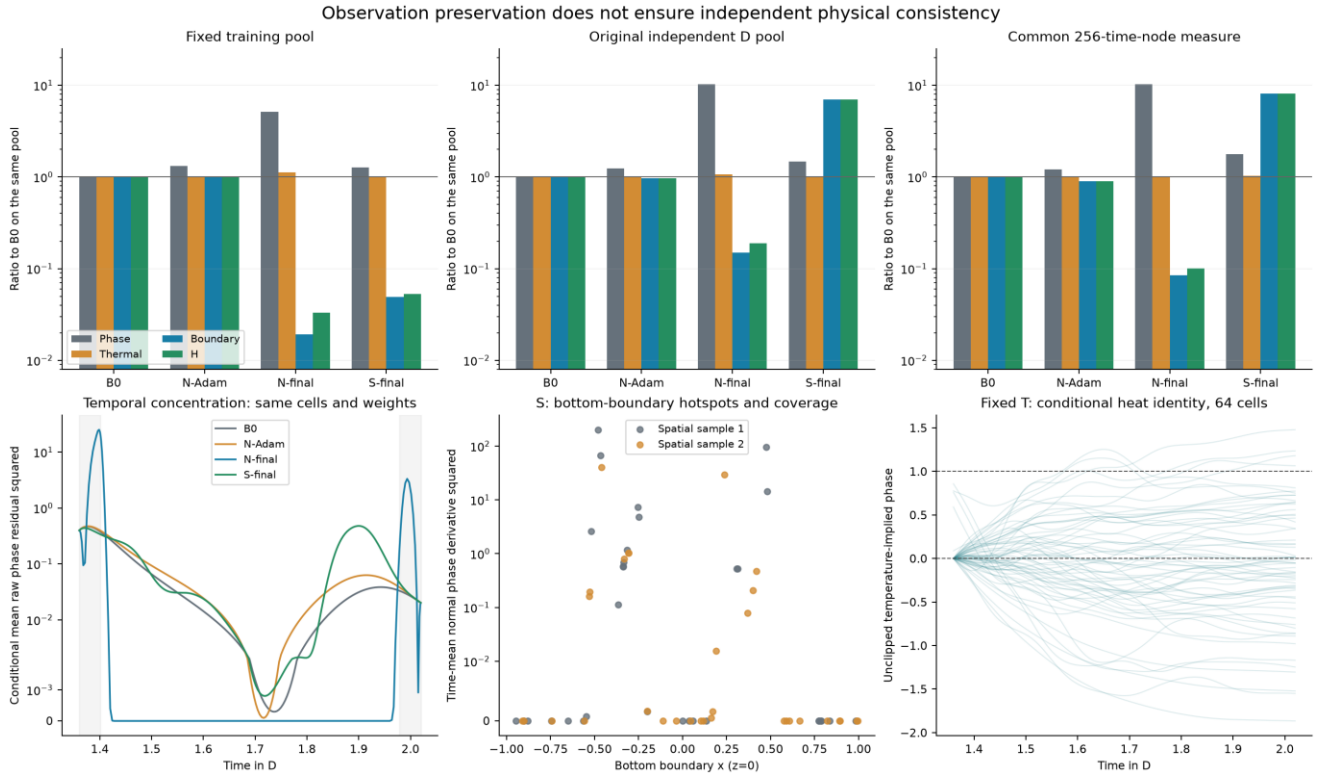
Two later development questions retain the same synthetic physical object but have distinct parents and budgets. An eight-arm relative-residual/time-moment study does not establish its prescribed phase-gap increment; its eight arms share one parent and are not eight independent initializations. The combined RIM candidate improves phase RMS by 1.918% and set error by 2.202% over its matched D control, below the inherited practical requirements. These effects must not be pooled with the earlier B1 comparisons or interpreted as a cross-round learning curve. Supplement S21 retains every arm, raw audit and event record.

A separate fixed-parent study restricts neural N and spline S corrections to the dark interval  $D=[1.36,2.02]$ , preserving the base temperature and port predictions and leaving phase unchanged outside D. General correction G is the direct unrestricted-support control. None establishes the prescribed completion increment or independent equation-wise qualification. N reduces the window set error by 34.21% while increasing phase RMS by 7.95%; unchanged ports therefore do not identify the hidden phase trajectory. These are changes in predictions, not new evidence of their accuracy.

For B0, the accepted N-Adam checkpoint, N-final and S-final, a subsequent zero-update diagnostic compares the same time nodes, spatial cells and boundary samples. At common 256-node temporal quadrature, N-final lowers H by 89.9073% while its raw phase-residual square becomes 10.3157 times B0. On a second spatial support these values are 80.7961% and 13.4327 times, respectively. The decomposition  $H=J_{\phi}/75+J_T/48+5 B_{\phi}$  shows compensation between boundary and dynamic terms. The tested 128-to-256 temporal refinement changes each integrated component by less than 0.029%, whereas the spline boundary term depends strongly on spatial coverage. This is a directly observed objective/qualification mismatch, not an isolated proof of the complete optimization failure mechanism or a general claim about L-BFGS.

The temperature-implied phase trajectory under exact thermal balance violates phase-range and endpoint conditions on the checked support. Its thermal-square lower bound is about 21.615% of the same-support base value, which does not exclude improvement within the allowed 105% thermal budget. The base itself belongs to that tolerance set. What remains unresolved is an attainable improvement satisfying the original support, range, endpoint and equation-wise conditions, followed separately by improved reference reconstruction and a neural-specific benefit. Supplement S22 preserves all controls and component records.

A supplementary fixed-temperature conditional evolution passes the two-step numerical checks and both paired thermal budgets. On the same 80 by 40 D-only development measure, phase RMS falls from 0.066944 to 0.020743, with the same direction in the original ROI. Its endpoint derivative and value mismatches remain substantial, so the result supports considering a bounded search within the original correction family without establishing that family's feasibility or a neural-specific increment (Supplement S23).



**Figure 6.** Saved-checkpoint diagnostic of scalar-objective reduction and separate physical components. Temporal refinement and spatial-support changes are distinct checks; values are not mixed across supports. This figure uses the completed zero-update diagnostic, not a new training run.

## 6. Physical interpretation and numerical scope

### 6.1 Pulse history and functional quantities

The references agree before the changed pulse to roundoff at matching resolution. Immediately before the second pulse, advancing it from 1.25 to 1.01 increases the reference ROI mean  $T$  from 0.006525 to 0.018746 and mean phase from 0.0004829 to 0.0039725. The second-event latency decreases from 0.2484 to 0.2168. These observations support history dependence in this numerical object but do not isolate thermal from phase mediation. Complete per-pulse states and event values remain in Supplement S5.

Device and event rankings differ. Historical shorter-protocol E passes both onset-timing tolerances under the original reference, but first-cycle recalls 0.866607 and 0.875507 remain below 0.9. F29 has the smaller second-cycle timing error. F43 has opposite signed pulse-energy errors, +0.00511674 and -0.00400099, yielding 0.25736% total energy error despite 3.09469% power-trajectory NRMSE. E also exhibits cancellation. Since projected current and power obey  $P=UI$ , their errors are related drive-weighted views; neither low energy error nor electrical balance certifies local heating or strict event reconstruction.

### 6.2 Reference and reader sensitivity

The original reference uses 160 by 80 cells and  $dt=0.000625$ ; temporal refinement halves  $dt$ ; spatial refinement uses 240 by 120 at the halved step. All save the same 1001 times and keep the learned states fixed. Spatial fields are conservatively restricted for the primary state/event measure while ports remain native. Predictions have two separately tested electrical readers. These are finite perturbation tests, not a demonstrated convergence order or continuum-error estimate.

Across the historical coarse-reader comparisons, temporal refinement changes no A/B decision; spatial refinement changes three: original/29 E/B\_E loses A, shorter/43 E/F loses A, and original/43 E/B\_E gains B. The latter B crossing disappears with the finer reader. Section 4.2 additionally retains the continuous B\_E reversals under original/time references. The complete E/F device criterion remains retained in every tested condition. Supplement S9 and S14-S16 give the complete margins, sufficient perturbation bounds and restriction conventions.

A later gated continuation crosses the strict rule only under the time-refined reference and loses it under spatial refinement as first-cycle recall changes from 0.902063024 to 0.897526906. Ordinary and gated phase-head

extensions do not establish a stable matched increment over unchanged continuation. All six negative continuation states are retained in Supplement S10-S13 and the separately indexed array extension. These limits prevent attributing the core effect to a demonstrated new gate, extra capacity or arbitrary-grid robustness.

### 6.3 What the experiment identifies

The E/F contrast identifies the tested constraint-training configuration. Initial V, active parameters, exact versus finite enforcement, enforcement times and gradient maps differ; the experiment does not isolate one implicit-gradient pathway. E/D\_E controls that distinction more tightly for the additional interior residuals, but its conclusion is conditional on the information regime and frozen optimization recipe. Each information regime rebuilds its own parents and calibration scales. The comparison between regimes therefore describes the observed pattern of increments without holding every training mediator fixed. Equal update limits do not give equal numerical work, and no acceleration over a conventional forward solver is claimed.

The domain has a localized electrode and a closed electric-thermal-phase feedback chain. Its dimensionless coefficients, single phase variable and idealized geometry still require a separate material-consistent calibration and independent validation before a device-material claim. The initial-boundary incompatibility and shared numerical discretization likewise remain limitations. A stronger algorithmic or material claim would need a separately designed experiment rather than relabeling the present controls.

## 7. Conclusions and availability

Training-time electrical elimination with consistent local Joule deposition produces a matched device-reconstruction benefit over the tested soft PINN after both receive the same final electrical repair. All four full-label pairs retain the complete device criterion under the declared reference and reader changes. Strong interpolation supplies a continuous reversal and a reader-sensitive advantage decision, limiting the scope of that benefit.

Matched full-label and phase-gap controls do not establish the expected additional dynamic-residual increment. The bounded relative-residual and observation-preserving studies retain this limitation; lower scalar loss and unchanged electrical ports do not establish a qualified internal phase reconstruction. Strict two-cycle usability remains unestablished. The evidence separates a configuration-level benefit from unresolved algorithmic, generalization and material claims.

### Data, code and author declarations

The curated repository through [commit e2e14b5](#) includes selected manuscript, B1, bounded-correction and diagnostic evidence. This integrated review version and its new conditional diagnostic are a subsequent local delivery. The full prediction/reference arrays remain local. The portable packages retain 72 core, 18 historical continuation and 42 B1 reference/reader records; these counts are not independent replications. The completed core arithmetic reproduction checked 8,868 scalar, identity and Boolean entries at  $\text{rtol}=2\text{e-}10$  and  $\text{atol}=2\text{e-}12$  with exact categorical agreement. Array rescoring, figure rebuilding, new model inference and retraining are distinct reproduction levels. No independent external retraining is claimed. The accompanying evidence map identifies the actual source records and figure inputs.

Full arrays have not been publicly uploaded or assigned a DOI. Public or controlled reviewer access, appropriate distribution permission and persistent archiving require author approval. AI-assisted preparation: OpenAI Codex assisted with research-code preparation, numerical execution, source checking, analysis, figures and drafting. Human authors must verify the content and take responsibility for interpretation, disclosures and submission. Funding, author contributions, competing interests, affiliations and final author approval remain to be supplied truthfully. This numerical study contains no experimental or human-subject data.

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